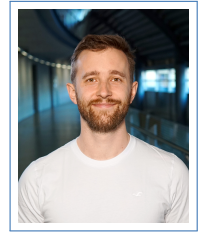


Lukas Burgholzer

Curriculum Vitae

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Lukas Burgholzer



Profile

Quantum-software researcher, engineering leader, and CTO working across the quantum computing software stack and HPCQC integration. I oversee the development of the *Munich Quantum Toolkit (MQT)* and the *Munich Quantum Software Stack (MQSS)*, including the Quantum Device Management Interface (QDMI), and co-founded MQSC. As of July 2026, my work has accumulated over **2.6k citations (h-index 30)**; MQT has over **4.1 million PyPI downloads** and **1.4k GitHub stars**.

Experience

- 2025–present **CTO and Founding Member of MQSC**, Bootstrapped deep-tech startup that translates quantum-software research into industrial products, <https://mq.sc>.
- 2024–present **Technical Lead of the Munich Quantum Software Stack (MQSS)**, Munich Quantum Valley initiative with 400+ researchers; lead developer and maintainer of QDMI, an emerging de facto standard for interoperable quantum-hardware integration in HPCQC software stacks, <https://github.com/Munich-Quantum-Software-Stack/QDMI>.
- 2023–present **Chief Developer of the Munich Quantum Toolkit (MQT)**, Open-source ecosystem providing tools across the quantum computing software stack, including simulation, compilation, and verification; 4.1M+ PyPI downloads and 1.4k+ GitHub stars (July 2026), <https://mqt.readthedocs.io>.
- 2023–present **Senior Researcher**, Technical University of Munich, Germany.
Chair for Design Automation. Mentorship of 8+ doctoral researchers; initiated and co-developed courses, adopted by DTU and Northwestern University.
- 2019–2023 **Doctoral Researcher**, Johannes Kepler University Linz, Austria.
LIT Secure and Correct Systems Lab. Institute for Integrated Circuits.
- 2018–2019 **Graduate Research Associate**, MathConsult GmbH, Linz, Austria.

Education

- 2019–2023 **PhD Studies in Computer Science**, Johannes Kepler University Linz, Austria.
Supervisor: Univ.-Prof. Dr. Robert Wille (Institute for Integrated Circuits), *passed with distinction*
- 2017–2018 **Master's Degree in Industrial Mathematics**, Johannes Kepler University Linz, Austria.
passed with distinction
- 2016–2019 **Bachelor's Degree in Computer Science**, Johannes Kepler University Linz, Austria.
passed with distinction
- 2013–2016 **Bachelor's Degree in Mathematics**, Johannes Kepler University Linz, Austria.
passed with distinction

Skills

- Languages German (native), English (fluent)
- Quantum software Tools across the quantum computing software stack: simulation, compilation, verification, optimization, mapping, and hybrid execution
- Algorithms Decision diagrams, automated and optimized compilation, and scalable verification
- Software engineering Modern C++, Python, Git, CMake, and CI/CD
- HPCQC and leadership HPCQC integration and QDMI; technical leadership for MQT, MQSS, and MQSC; mentorship of doctoral researchers

Awards and Selected Achievements

- 2025 EDAA Outstanding Dissertation Award, European Design and Automation Association.
- 2024 ACM SIGDA Outstanding PhD Dissertation Award.
- 2024 GI Dissertation Award nomination, Gesellschaft für Informatik, Austrian Computer Society, and Swiss Informatics Society.
- 2024 Heinz Zemanek Prize, Austrian Computer Society.
- 2024 PhD Forum Best Poster Prize, Design, Automation and Test in Europe (DATE).
- 2023 Best Paper Award nomination, Asia and South Pacific Design Automation Conference.
- 2022, 2023 Mentor, NYUAD International Hackathon for Social Good in the Arab World (quantum computing).
- 2022 JKU Young Researcher's Award, Johannes Kepler University Linz.

Research and Publications

Quantum computers are rapidly advancing, yet the corresponding software lags behind. Through my research, I bridge this gap by transferring methods from classical design automation to the quantum realm—creating the essential software layer that makes quantum hardware usable. My contributions include decision-diagram-based simulation methods that reduce workloads from days to minutes, automated and optimized compilation methods for diverse hardware platforms, compiler infrastructure based on MLIR, and efficient, scalable verification. I oversee the development of the *Munich Quantum Toolkit (MQT)* and the *Munich Quantum Software Stack (MQSS)*, including QDMI—an emerging de facto standard for interoperable quantum-hardware integration—ensuring that powerful quantum processors can be integrated into high-performance computing centers and utilized by end-users. My work covers the entire quantum computing software stack: from efficient data structures and simulation to compilation, verification, and hybrid execution.

I have published 80+ papers in leading venues including IEEE TCAD, ACM TQC, DAC, DATE, ICCAD, QCE, and SC. For a full list of publications, see <https://www.cda.cit.tum.de/team/burgholzer/>.

Open-Source Software

I am a strong advocate for open science. All methods developed in my research are released as part of the *Munich Quantum Toolkit (MQT)* (<https://github.com/munich-quantum-toolkit>), forming a comprehensive, open-source ecosystem for quantum software. These tools are engineered for performance (C++) and usability (Python), enabling thousands of researchers and developers worldwide to build quantum applications. The MQT has become a staple in the community, with over **4.1 million downloads** on PyPI and more than **1.4k stars** on GitHub (as of July 2026). Tools natively integrate with IBM Qiskit, Xanadu's PennyLane, and `jeff`; they are deployed at HPC centers across Europe and beyond, including the Leibniz Supercomputing Centre (LRZ) and Oak Ridge National Laboratory, and serve as the technological backbone of MQSC.

For a full list of contributions, see <https://github.com/burgholzer>.